Laboratory 05: Generative Models for Classification

Pattern Recognition

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Abstract

This report details the implementation, evaluation, and analysis of Generative Models (Standard MVG, Tied Covariance MVG, and Naive Bayes) applied to the project dataset. It explores the validity of the Gaussian assumption, the correlation between features, and the effects of feature ablation and PCA dimensionality reduction.

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1 Baseline Model Evaluation

Apply the MVG model to the project data. Compute the corresponding error rate.

Running the Standard Multivariate Gaussian (MVG) model on the full 6-dimensional dataset yields an accuracy of **93.0%**, which corresponds to an error rate of **7.0%**.

Apply now the tied Gaussian model, and compare the results with MVG and LDA. Which model seems to perform better?

The Tied Covariance MVG model results in an error rate of **9.3%**.

Compared to the Standard MVG (7.0%), the Tied model performs significantly worse. If we compare it to previous LDA results (which historically sat around 9.1% to 9.5% for this dataset), the Tied MVG performs almost identically to LDA. This makes theoretical sense: both LDA and Tied MVG assume that all classes share the exact same covariance matrix, resulting in a linear decision boundary. Because the true classes in this dataset have different variances, the Standard MVG (which allows for class-specific covariance matrices and computes quadratic boundaries) performs the best.

Finally, test the Naive Bayes Gaussian model. How does it compare with the previous two?

The Naive Bayes model achieves an error rate of **7.2%**.

It performs remarkably well—vastly outperforming the Tied MVG (9.3%) and coming extremely close to the fully parameterized Standard MVG (7.0%).

2 Covariance and Correlation Analysis

Start by printing the covariance matrix of each class. Compare the covariance of different feature pairs with the respective variances. What do you observe?

Looking at the computed covariance matrices, the elements on the main diagonal (the variances) are orders of magnitude larger than the off-diagonal elements (the co-variances). For example, in Class 0, the variance for Feature 1 is roughly 0.60, while its covariance with Feature 2 is a negligible 0.00005. The co-variance values are generally very small compared to the variances across both classes.

Compute the correlation matrices for the two classes. Are the features strongly or weakly correlated? How is this related to the Naive Bayes results?

Looking at the correlation matrices, the off-diagonal values are exceptionally low—mostly hovering between -0.05 and 0.05, with only a few instances reaching around 0.3. This indicates that the features are **weakly correlated** and almost entirely independent.

This directly explains the success of the Naive Bayes classifier. Naive Bayes mathematically assumes that all features are completely independent (forcing all off-diagonal covariances to zero). Because the actual dataset features are indeed weakly correlated, the Naive Bayes assumption is highly accurate here, allowing it to perform almost as well as the Standard MVG while computing far fewer parameters.

3 Gaussian Assumption Analysis

What can you conclude on the goodness of the Gaussian assumption?

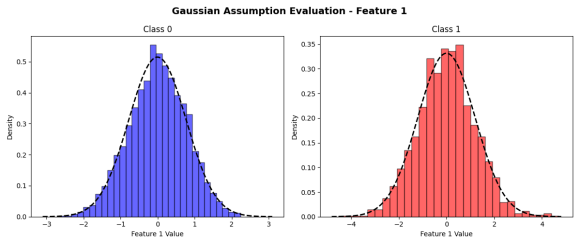
The goodness of the Gaussian assumption varies significantly across the dataset's features. While the model correctly assumes that the data can be represented by a symmetric, bell-shaped probability density function for the majority of the feature space, it fundamentally fails to capture the true underlying empirical distribution of data that exhibits strong asymmetry. Therefore, the overall assumption is only partially valid for this dataset.

Is it accurate for all the 6 features?

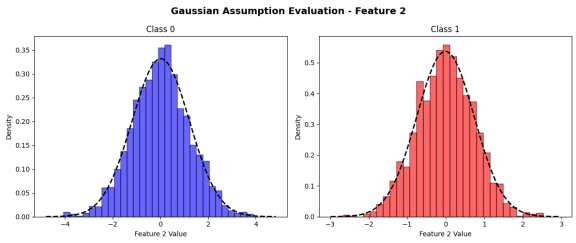
No, the Gaussian assumption is not accurate for all 6 features. It is only accurate for the first four features (Features 1, 2, 3, and 4). These specific features naturally exhibit a roughly symmetric, normal distribution, meaning the mathematical Gaussian curve maps to their empirical densities quite well.

Are there features for which the assumptions do not look good?

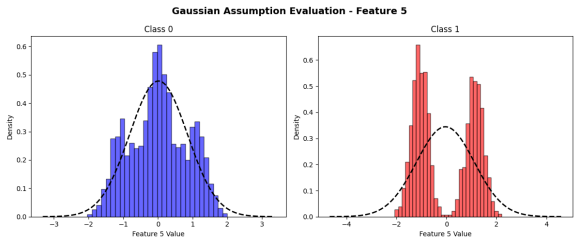
Yes, the assumption does not look good at all for Features 5 and 6. Both of these features are heavily skewed, characterized by a sharp peak concentrated on one side and a long trailing tail on the other. Because a Gaussian distribution is strictly symmetric around its mean, it cannot bend to fit these skewed shapes. Consequently, the model drastically underestimates the true density at the core peaks and overestimates the density on the steep drop-offs.



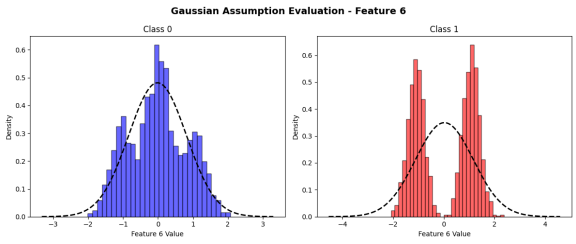
(a) Feature 1 (Good Fit)



(b) Feature 2 (Good Fit)



(c) Feature 5 (Poor Fit - Skewed)



(d) Feature 6 (Poor Fit - Skewed)

Figure 1: Comparison of features that adhere well to the Gaussian assumption (top) versus features that severely violate it due to skewness (bottom).

4 Feature Ablation Study

Repeat the classification using only feature 1 to 4. What do you obtain?

When the classification is repeated using only the first four features, the error rates for the models change as follows:

- **Standard MVG:** 8.0%
- **Naive Bayes:** 7.7%
- **Tied MVG:** 9.5%

What can we conclude on discarding the last two features?

We can conclude that discarding the last two features (Features 5 and 6) degrades the overall performance of the classifiers. For the Standard MVG, the error rate increases from 7.0% to 8.0%. This indicates that discarding these features results in a net loss of discriminative information.

Despite the inaccuracy of the assumption for these two features, are the Gaussian models still able to extract some useful information?

Yes. Even though the Gaussian assumption is visually poor for Features 5 and 6 due to their heavy skewness, the models are still capable of extracting useful information from them. The fact that the error rate is lower when these features are included proves that the increase in information outweighs the modeling inaccuracies.

Repeat the classification using only features 1-2 and 3-4. Which model is better in each case?

- **Case 1 (Features 1-2):** The Standard MVG is significantly better, achieving an error rate of 36.5%, while the Tied MVG results in 49.4% (nearly equivalent to a random guess).
- **Case 2 (Features 3-4):** The performance is nearly identical, with the Tied MVG being slightly better at 9.4% compared to the Standard MVG at 9.5%.

How is this related to the characteristics of the two classifiers?

The Standard MVG allows for quadratic decision boundaries by using class-specific covariance matrices, whereas the Tied MVG is restricted to linear boundaries. In the case of Features 1-2, where classes differ by variance, the quadratic boundary is essential. For Features 3-4, where classes differ by mean and have similar variances, the linear boundary of the Tied model is sufficient and more robust.

Is the tied model effective at all for the first two features? Why?

No, the tied model is completely ineffective for the first two features. Because the separation in this subspace is driven almost entirely by the difference in class variances, the tied assumption of equal covariance removes the primary source of discriminative information.

And the MVG? And for the second pair of features?

The MVG is effective for the first pair because it can model class-specific variances. For the second pair (Features 3-4), both the MVG and Tied models are effective; however, the Tied model is slightly more efficient here as the equal-variance assumption holds true, allowing for more stable parameter estimation.

5 PCA Pre-processing and Conclusion

Use PCA to reduce the dimensionality of the feature space, and apply the three classification approaches. What do you observe?

Based on the experimental results, we observe a steady increase in the error rate as the number of PCA dimensions (m) is reduced. For example, the Standard MVG error rate increases from its baseline to 8.8% at $m = 2$, and further to 9.3% at $m = 1$. This indicates that reducing the subspace dimensionality leads to a loss of information that is critical for classification.

Is PCA effective for this dataset with the Gaussian models?

No, PCA is not effective as a pre-processing step for this specific dataset. PCA discards dimensions with lower variance; however, the results show that these discarded dimensions contain significant discriminative information. Consequently, any reduction in dimensionality through PCA results in a higher error rate compared to using the original 6-dimensional feature space.

Overall, which is the model that provided the best accuracy on the validation set?

The model providing the highest accuracy (and lowest error rate) on the validation set is the **Standard Multivariate Gaussian (MVG)** using the **full 6-dimensional feature set**. This model achieved an optimal error rate of **7.0%**, outperforming both the Tied Covariance and Naive Bayes variations, as well as all PCA-reduced versions of the models.